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PODNEBNI SKLAD

Quality Assessment of Surface Current and Wave Measurements from the High-Frequency Radar System in the Gulf of Trieste

Section 3:

Development of new quality-control (QC) algorithms for the detection of anomalies that remain undetected by existing algorithms

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1 Introduction

Based on the findings of the Work Project 2: Evaluation of the effectiveness of existing quality control (QC) algorithms for measurement data, hereafter *wp2*, the objective of this work package is to develop or adapt new procedures for the detection and flagging of anomalies and artefacts in the HFR GOT for the 2021–2026 time series, that are not addressed by the current algorithms.
In particular, the new QC algorithms should be capable of detecting non-physical divergences and convergences in the flow field. The algorithms should enable reproducible processing and the documented removal or flagging of suspicious values. It should also be investigated whether unsupervised artificial intelligence algorithms can be used to distinguish clusters of non-physical artefacts from other situations.

Note 1: All scripts and corresponding results are provided alongside this document.

2 Selection of the types of anomalies/artefacts to be addressed using new algorithms, based on the report from WP2

The HFR-NAdr network publishes data through the EU HFR Node, which applies a standardized QC suite to all member radar systems before publication to ERDDAP. QC is applied at two product levels: *Radial product*: (per-station: AURI, IZOL, PIRA, TRI1), and *Total product*. A vector can therefore be flagged at the radial stage, the total-combination stage, or both.

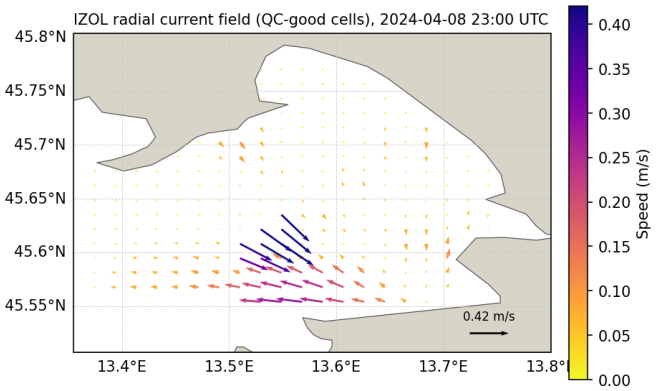


Figure 2: Velocity field example from HFR-NAdr-IZOL at 2024-04-08 23:00UTC, where all QC algorithms passed with QCflag=1.

In wp2, tasks 2 and 3 established two different failure modes for the existing QC:

- Where an existing test is well-matched to a real problem, RDCT_QC against station outages and low radial count, GDOP_QC against the AURI/PIRA baseline geometry, CSPD_QC/VART_QC/MDFL_QC against extreme or unstable individual vectors.
- Where the failure has no matching existing test, it passes as "good data" regardless of how implausible it looks. Figure 2 is a concrete example.

Overall, the existing EU HFR Node QC does its intended job well within its own scope, i.e. geometry and count-based rejection. However, it was not designed to detect physical-consistency issues, including non-physical divergence, direction reversals, and inconsistencies between neighboring vectors, which are the focus of this work.
The new algorithms should detect unphysical divergences/convergences (two small opposing patches side by side, or a patch flowing against its surroundings), unphysical shear (a narrow strip flowing to both sides, direction flipping between adjacent cells), single-cell reversals/spikes (one cell opposite to or much stronger than all its neighbours), and ideally Bora or Scirocco outbursts (strong coherent jets with a hard edge). Strong outbursts are physically real, so they are not treated as errors.

3 Development or adaptation of QC algorithms for the selected types of anomalies. Documentation of the methods and parameters used

Note 2: No new data were downloaded; this analysis was performed entirely on the already unified and published archive. All five products share one regular 20×22 lat/lon grid (1.5 km) and the same half-hour time steps.

For this analysis, two algorithms were developed and tested to detect the selected types of anomalies: k-means clustering, and convolution with different types of kernels and a sliding window.

In both analyses, a cell's value was considered valid only if its QC flag was good (flag 1 or 2 in Table 3, wp2) of the existing EU HFR Node QC. Cells with bad QC flags (3 or 4) were treated identically to true no-coverage NaN values and were therefore not used as input. The new flags look for errors the existing QC lets through, so they add to the existing QC and do not replace it.

3.1 K-means clustering

For this analysis, the script *cluster_regimes.py* was developed, while the outputs are `cluster_summary_<product>.csv` and `cluster_<product>_k<k>_cluster<c>.png`.

The script ran k-means, with k up to 8, over the full 2021-present HFR archive: once for the Total product (one feature vector per timestep, built jointly from EWCT(u)+NSCT(v)), and once for each radial station (AURI, PIRA, TRI1, IZOL; one feature vector per timestep per station, built from RDVA only). It then rendered the mean field of each cluster, so that it could be assessed whether the clusters corresponded to recognizable circulation regimes (Bora, Scirocco, Tramontana) and/or whether a divergence-dominated cluster emerged.

The results were not satisfactory with respect to the intended objective. For example, for the Total product, for $k = 2$, the clustering primarily separated the observations according to flow direction, resulting in two clusters corresponding to a northeastward (NE) and a southwestward (SW) flow, respectively (see example in Figure 3 (left)). However, increasing the number of clusters did not lead to the identification of additional clusters representing distinct flow patterns or behaviours. Instead, the algorithm continued to subdivide flows exhibiting the same direction according to their magnitude. Consequently, the initial separation into NE and SW-directed flows progressively evolved into clusters representing strong and weak SW flow, as well as strong and weak NE flow (see example for $k = 6$ in Figure 3 (right)). This behaviour was consistently observed also for the Radial products.

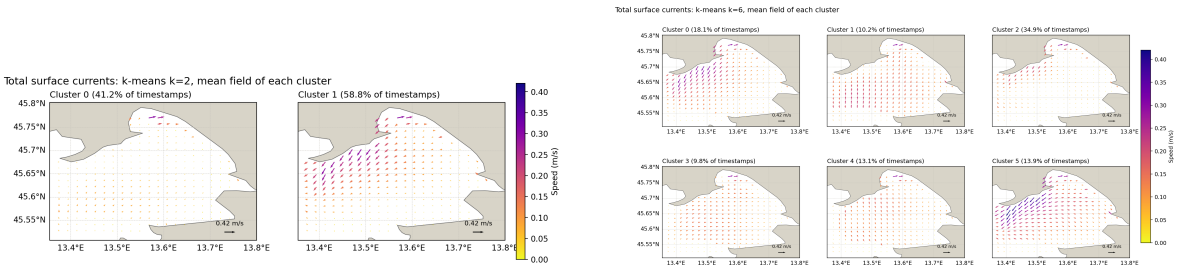


Figure 3: k-means clustering with k=2 (left) and k=6 (right) on Total surface current field.

The key result was that the divergence events disappear in cluster-mean fields, because they occur at a different place each time. The main limitation can be attributed to the averaging: each cluster is represented by the average field of all the time points within it, whereas divergences occur at different locations and times, so they cancel each other out in the average.

k-means with k=10, 15, 20, 25
The analysis used the same script *cluster_regimes.py* and the same input as for k = 2-8: whole-field snapshots of the time steps with at least 75% of the product's typical coverage, EWCT (u) and NSCT (v) for the total product and RDVA for the radial products. The outputs are `cluster_summary_<product>_k10-25.csv`, `kmeans_reference_lookup_k10-25.csv`.

For the total product, the mean fields of all clusters were smooth variants of the known regimes: more clusters only cut the data into finer slices. Higher k mainly added intensity steps of the same regimes (Bora, Scirocco, and Tramontana). None of the cluster mean fields shows a divergence or convergence line (see example below in Figure 4, for k=20).

For each reference case (Table 2), the cluster and the cluster size were looked up. Most reference cases never reached the clustering, because their coverage was below the 75% threshold. Artefacts tend to occur in partly covered time steps, near the edge of the coverage, while whole-field clustering only uses well-covered ones.

To let more reference cases into the clustering, the analysis was repeated with a coverage threshold of 60% and with cells that are valid in less than 10% of the time steps removed. The total product then included 7 of its 10 reference cases, but they ended up in the largest clusters, which collect the partly covered time steps. k-means therefore grouped them by how much of the map was missing, not by the divergence.

Given that this analysis did not render the expected results, it was not pursued further, and an alternative approach was explored.

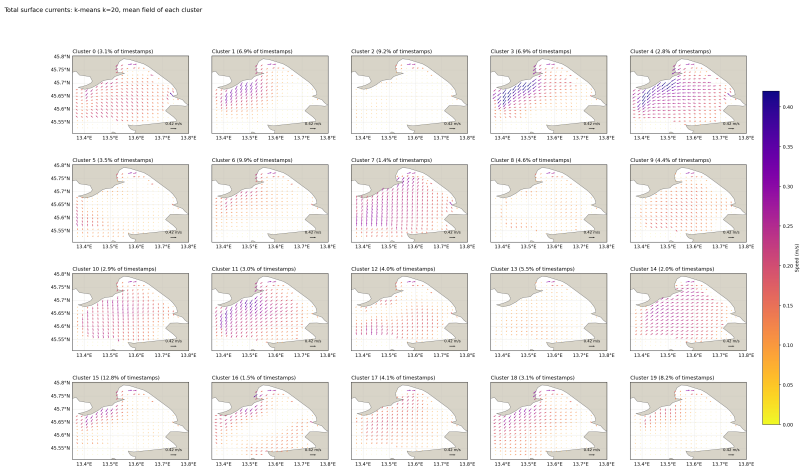


Figure 4: k-means clustering with k=20 on Total surface current field.

3.2 Sliding-window kernels + Isolation Forest

For this analysis, the scripts *kernel_features.py*, and *kernel_clustering.py* were developed, alongside the outputs `kernel_anomaly_map_<product>[_3x3].png`, `kernel_anomaly_monthly_<product>[_3x3].png`, `kernel_frequency_<product>[_3x3].png`, and `kernel_featuremaps_<product>_<time>.png`.

Unphysical divergence, convergence and shear appear as small, local features, a few neighbouring cells whose velocities point in opposite directions or change abruptly over one grid cell. Because these features occur at a different place each time, they average out when the whole current field is analysed at once. Therefore, the field was analysed locally, using a sliding window: a small square of 3×3 or 4×4 grid cells that is moved across the whole grid one cell at a time, at every half-hourly time step. Each window position is treated as a separate sample.
A window is only used when all of its cells are marked as good by the existing QC, so the method works only on data that the current QC already accepts.

Within each window four *half-split kernels* were computed (see Table 1 and Figure 5). A kernel is a fixed pattern of weights that divides the window into two halves, A and B, and returns the mean velocity of half A minus the mean velocity of half B.
The four kernels compare north with south (TB), east with west (LR), and the two diagonal halves (DR and UR).

Kernel	**Half A**	**Half B**	**4 x 4 cells**	**3 x 3 cells**
TB (top/bottom)	north rows	south rows	8 / 8	3 / 3 (middle row weight 0)
LR (left/right)	east columns	west columns	8 / 8	3 / 3 (middle column weight 0)
DR (diagonal ↘)	NE triangle	SW triangle	6 / 6 (diagonal 0)	3 / 3 (diagonal 0)
UR (diagonal ↗)	NW triangle	SE triangle	6 / 6 (diagonal 0)	3 / 3 (diagonal 0)

Table 1: Definition of the kernel configurations and corresponding cell groups.

For the total product the kernels were applied separately to the eastward EWCT (u) and northward NSCT (v) components, which gave eight values per window. For the radial products they were applied to the radial velocity, which gave four values.

The kernel values have a direct physical meaning. A large positive LR value on u or TB value on v means that the two halves move apart (divergence), and a large negative value means that they move towards each other (convergence). Large LR values on v or TB values on u indicate shear. A smooth current field gives small kernel values, while a sud-

den jump or reversal between neighbouring cells gives a large value in at least one kernel.

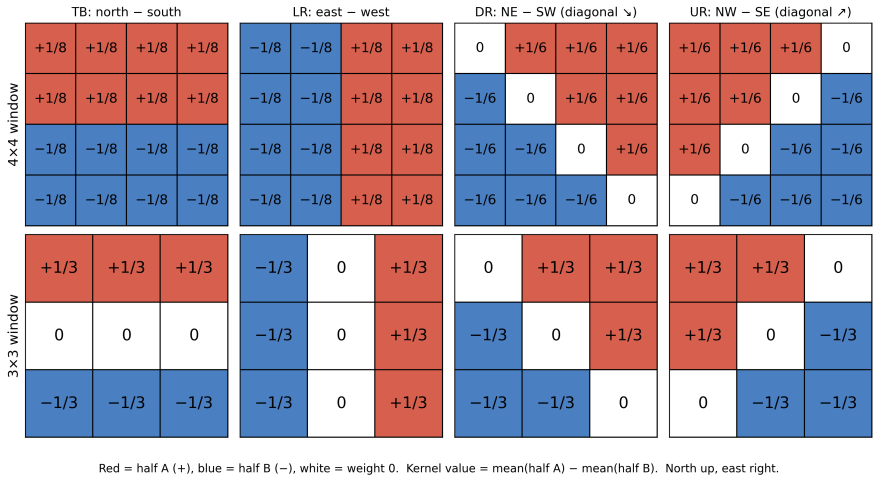


Figure 5: The four half-split kernels for the 4×4 (top) and 3×3 (bottom) windows. Each kernel returns the mean of half A (red, +) minus the mean of half B (blue, −); white cells have weight 0.

The *Isolation Forest* algorithm, an unsupervised anomaly-detection algorithm that needs no labelled examples, was used to find the windows whose combination of kernel values is unusual. The algorithm builds an ensemble of random binary trees: at each step it picks a random feature (here, a kernel value) and a random split value, and keeps dividing the data until each sample is isolated. Unusual samples lie far from the bulk of the data, so they are separated after only a few splits, whereas typical samples need many. The average number of splits over all trees is converted into an anomaly score, where a higher score means a more unusual window. The forest (200 trees) was trained on a random sample of up to 500,000 windows per product and then used to score every window in the 2021-2026 archive.

To make the selection concrete, the most anomalous events found by the sliding-window kernels and the Isolation Forest were reviewed visually. For each product, the five highest-scoring events for each window size (4×4 and 3×3) were selected, at least three days apart, giving 50 events in total. For every event, the current field was inspected together with the actual velocity values read from the netCDF files. Each event was classified as an error, a physical feature, or uncertain.
From this review, 31 reference cases were selected (Table 2). In 26 of them the field contains one of the target artefacts, although the existing QC algorithms rate all the cells involved as good.
The objective was for the new algorithm to flag these cases. The remaining cases were physical features or not artefacts, and the new algorithm must not flag them. These 31 cases were later used to tune and test the new flags.

Product	Divergence/convergence	Shear	Other anomalies
Must be flagged			
Total	2023-02-02 20:00, 2023-12-28 01:00, 2024-10-22 08:00, 2025-11-08 07:00	2024-02-02 08:00, 2023-12-31 04:30	2023-11-11 10:00*, 2023-06-21 07:00*
AURI	–	2022-03-29 11:30, 2026-02-25 01:00	2022-01-02 11:00
PIRA	2022-02-23 17:00, 2023-04-23 05:00, 2024-08-08 01:30	2025-07-09 20:30, 2022-03-23 07:30	2024-12-05 10:30
TRI1	2024-10-22 03:30, 2024-03-15 08:30	–	2024-06-11 17:00, 2024-03-03 19:30*
IZOL	2024-06-28 08:00, 2023-06-28 22:00, 2023-11-01 22:00, 2024-07-08 03:30	–	2024-04-08 23:00
Must not be flagged			
PIRA 2022-06-24 10:30; TRI1 2023-10-02 04:30; TRI1 2023-11-03 04:00 (front also seen by AURI and IZOL); Total 2024-10-01 17:00 (Bora jet); Total 2025-12-25, whole day (Bora storm).			

Table 2: Reference cases selected from the visual review (times in UTC).
* single-cell spike or reversal.

The following examples (Figures 6 - 10) illustrate the results of the aforementioned evaluation for both the 4×4 and 3×3 windows. One example is presented for each product, selected from one of the days listed in Table 2. Overall, the 3×3 windows performed better.

All figures follow the same layout. For the radial products, the left-hand panel shows the QC-good radial velocity (RDVA) field and the corresponding radial velocity vectors. The pink box indicates the most anomalous 4×4 window, while the cyan box indicates the most anomalous 3×3 window, as identified by the Isolation Forest score. The right-hand panels show the values of the four kernels (rows: TB, LR, DR, and UR) applied to RDVA at each window position, first for the 4×4 windows and then for the 3×3 windows. Each kernel value is calculated as the difference between the mean radial velocities of the $+$ and $-$ halves (m/s). White indicates agreement between the two halves, whereas dark red and dark blue indicate large differences in radial velocity between neighbouring cells.
For the Total product, the left-hand panel shows the QC-good current velocity field. The pink and cyan boxes indicate the most anomalous 4×4 and 3×3 windows, respectively, as identified by the Isolation Forest score. The right-hand panels show the values of the four kernels (rows: TB, LR, DR, and UR) applied separately to the zonal (u) and meridional (v) velocity components at each window position, first for the 4×4 windows and then for the 3×3 windows. Each kernel value is calculated as the difference between the mean velocities of the $+$ and $-$ halves (m/s). White indicates agreement between the two halves,

while red and blue indicate that the + half has a higher or lower velocity component, respectively, than the − half. Specifically, red values for the TB kernel applied to v and the LR kernel applied to u indicate divergence, whereas blue values indicate convergence.

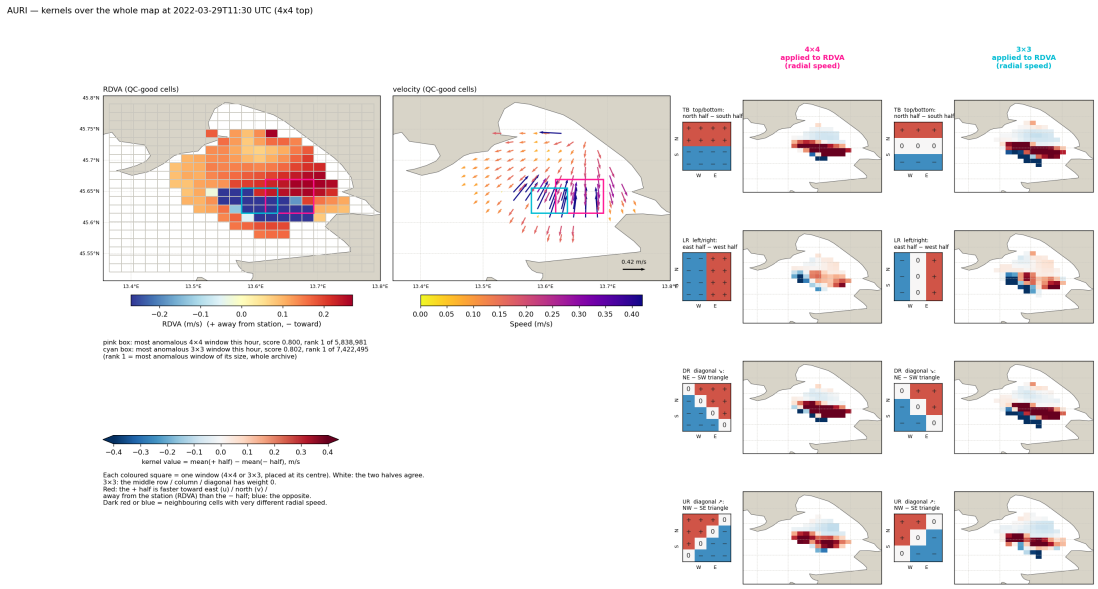


Figure 6: Kernel feature maps for the AURI radial product on 2022-03-29.

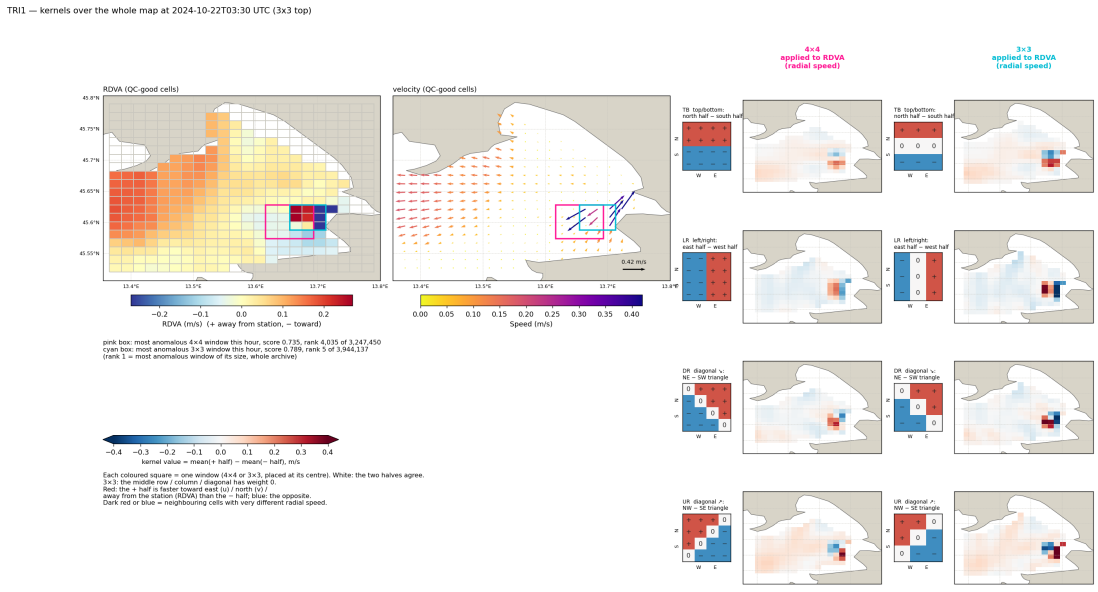


Figure 7: Kernel feature maps for the TRI1 radial product on 2024-10-22.

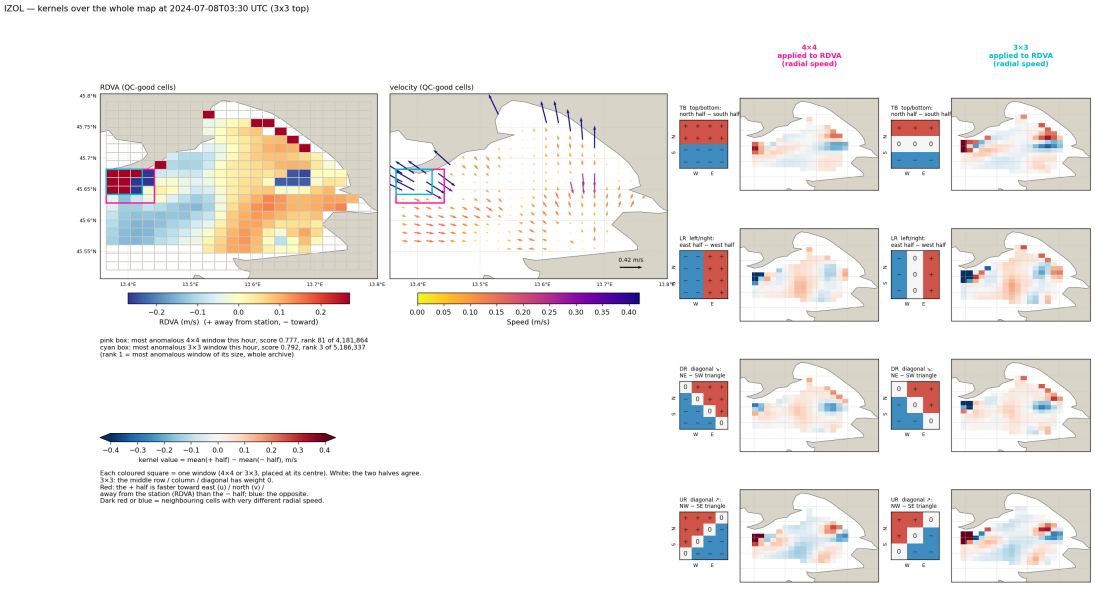


Figure 8: Kernel feature maps for the IZOL radial product on 2024-07-08.

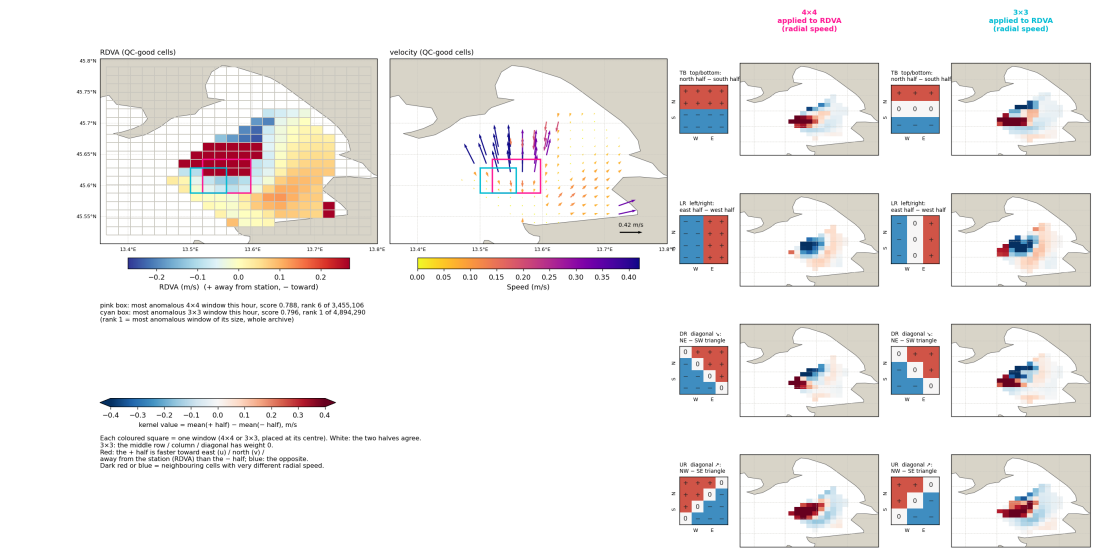


Figure 9: Kernel feature maps for the PIRA radial product on 2024-08-08.

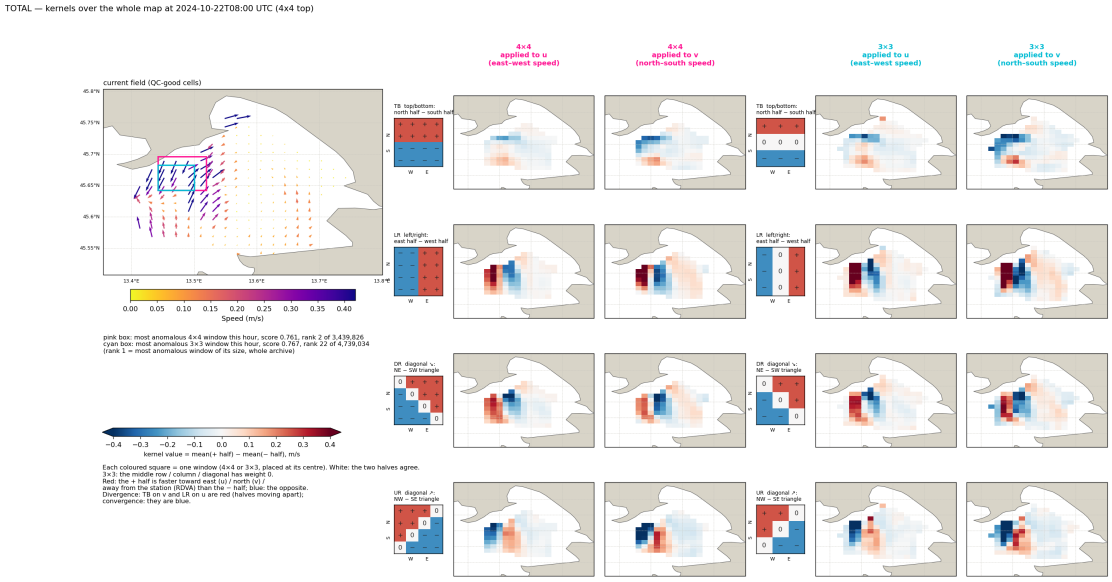


Figure 10: Kernel feature maps for the Total product on 2024-10-22.

Kernel-based QC flag

The anomaly score of the Isolation Forest identifies unusual windows, but an unusual window is not necessarily an error: a strong Bora jet next to calmer water is also unusual, yet physically real.

To turn the kernel results into a quality-control test, a flag that detects unusual windows, sorts them by type, and checks them against the other stations was built on top of them. The flag uses only cells marked as good by the existing EU HFR Node QC. It therefore complements the existing QC rather than replacing it.

Design choices

The design of the flag was guided by three preliminary tests on the reference cases selected (Table 2).

Firstly, two ways of selecting suspicious windows were compared: a physical threshold on the largest absolute kernel value in the window (criterion A) and a threshold on the Isolation Forest score (criterion B), both set at the same top fraction of the archive. For the radial products the two criteria selected the same windows in 87–89% of cases. For the total product they agreed in only 49% of cases, because criterion A selects windows with one very large jump, while criterion B selects windows that are moderately unusual in several kernels at once. At the top 0.5%, criterion A detected all reference cases and criterion B detected 23. A window is selected if it satisfies criterion A or criterion B.

Secondly, a reversal (neighbouring cells flowing in opposite directions, the target type) and an outburst (a region of strong flow next to weaker flow in the same direction) needed to be distinguished. Comparing the means of the two halves of the window did not work for the radial products, because a single opposite cell does not change the sign

of a three-cell mean. Comparing every pair of neighbouring cells inside the window did work: a single pair of cells with opposite directions is enough to mark the window as a reversal. With this rule, the reference artefacts were classified as reversals, while strong but coherent flows, such as the Bora cases, were classified as outbursts.
Lastly, the third test examined whether a radial reversal can be confirmed by the other stations. Because all five products share the same grid and the same half-hourly time steps, the same location can be checked in the other radial products at the same time.

Flagging rule
In the first step, a complete 3×3 window (all nine cells good) is selected as suspicious if its largest absolute kernel value exceeds the threshold thr_A or its Isolation Forest score exceeds the threshold thr_B. Both thresholds are defined as the top 0.5% of the corresponding values over the whole 2021-2026 archive of each product.
In the second step, each selected window is classified as a reversal or an outburst. A window is a reversal if at least one pair of neighbouring cells points in opposite directions, i.e. the angle between the two velocity vectors exceeds 90° (total product) or the two radial velocities have opposite signs (radial products). In addition, both cells must have a speed of at least $s_{\min}$, and their difference must be at least 0.3 m/s. All other selected windows are classified as outbursts. Outbursts are not treated as errors and remain flagged as good.
In the third step, each reversal in a radial product is checked against the other three radial stations within the window extended by one cell on each side (5×5 region) at the same time step. A station shows a jump if two neighbouring cells inside this region differ by at least 0.3 m/s. If at least two of the other stations show a jump, the reversal is considered a real feature and the cells remain good. If none of them shows a jump and at least one station has a smooth field there (at least five good cells), the reversal is contradicted by the independent observations and the cells are flagged as bad (flag 4 in Table 3, wp2). In all other cases, i.e. when only one other station shows a jump or no other station has data, the cells are flagged as probably bad (flag 3).
In the fourth step, reversals in the total product are always flagged as bad (flag 4). The total product is computed from the radial velocities, so the radial stations cannot confirm a total-product feature independently. Instead, the radial stations that show a jump at that location are recorded as the probable source of the reversal.
Finally, all nine cells of a flagged window receive the result of that window. When a cell belongs to several flagged windows, the most severe flag is kept.

Output variables
The flag produces three variables per grid cell and time step (Table 3). `KREV_QC` is the QC itself and follows the same convention as the existing EU HFR Node QC flags. `KOUT_FLAG` marks outbursts and confirmed real features. It is not a quality rejection; it is intended for a possible future climatological analysis, where such strong events can be studied separately. `KREV_SRC` is defined only for the total product and records which radial station or stations show a jump at the location of a total-product reversal.

Variable	Values	Meaning
KREV_QC	0, 1, 3, 4	0 = not tested (cell not inside any complete 3×3 window), 1 = good, 3 = probably bad, 4 = bad
KOUT_FLAG	0, 1	1 = outburst or reversal confirmed by at least two other stations (informational)
KREV_SRC	station list	total product only: radial station(s) showing a jump at a total-product reversal

Table 3: Output variables of the kernel-based QC flag.

Parameters

The parameters of the flag are listed in Table 4. They were first set from the preliminary tests and then tuned on the reference cases.

Parameter	Value	Meaning
Top fraction	0.5%	thr_A and thr_B are set at the top 0.5% of each product's archive
$s_{\min}$, total	0.15 m/s	Minimum speed of both cells in a reversal pair
$s_{\min}$, radials	0.05 m/s	Minimum absolute radial velocity of both cells in a reversal pair
Angle	90°	Total product: two vectors are opposite above this angle
Minimum difference	0.3 m/s	Minimum velocity difference within a reversal pair
Jump at other stations	0.3 m/s	Minimum difference between neighbouring cells at another station
Minimum cells	5	Number of good cells required at another station in the 5×5 region for it to count as smooth
Extension	1 cell	The 3×3 window is extended to a 5×5 region for the cross-station check

Table 4: Final parameters of the kernel-based QC flag.

4 Validation of the new algorithms on representative sections of the 2021–2026 time series and adjustment of the parameters

For this analysis, the scripts *kernel_qc_flag.py*, *kernel_qc_report.py*, and *plot_qc_flagged_field.py* were developed.

Validation design
The new method was validated at three levels (Table 5).
First, it was checked on the whole 2021–2026 archive that the kernels and the Isolation Forest score measure what they are intended to measure.
Second, the flag was tuned and tested on the 31 reference cases (Table 2).
Third, the flag was applied to three complete months, chosen to represent different conditions, in order to assess its behaviour independently of the reference cases. The flag was computed for every day containing a reference case and for the three validation months.

Part	Data	Purpose
Kernels and anomaly score	Whole 2021–2026 archive, all five products	Check that the features and the score detect the intended anomalies
Reference cases	31 cases (26 must be flagged, 5 must not)	Tuning of the parameters and testing against labelled events
Validation months	December 2025 (winter, Bora); July 2026 (calm summer, all four stations); October 2024 (most anomalous month of the archive)	Assess behaviour under different conditions, independently of the reference cases

Table 5: Validation design.

Validation of the kernels and of the anomaly score
The most anomalous windows coincide with the anomalies targeted in this work.
In the radial products, the top 1% contain a pair of adjacent cells with opposite radial velocity 18 to 49 times more often than the average window. Strong but uniform flow is not considered anomalous.
The 3×3 windows performed better than the 4×4 windows in every check. They cover 30–40% more of the archive, because fewer cells have to be good at the same time. For this reason the QC flag uses 3×3 windows only.
In the total product, the anomalous windows are concentrated at the western edge of the coverage (13.37–13.42°E, 45.63–45.68°N), where about 3% of the windows are in the top 1%, while the open interior of the gulf stays at or below 1%. The same position is the strongest hot spot of IZOL station.

Parameter tuning
Each parameter of the flag was varied separately around its starting value, while the oth-

ers were kept fixed (Table 6). For each setting, the following were counted: how many of the 26 cases that must be flagged received flag 3 or 4, how many of the 5 cases that must not be flagged remained good, and the share of tested cells flagged 3 or 4 in the three validation months. With the starting values, only the Christmas 2025 Bora day was handled incorrectly: two of its windows were classified as reversals with a minimum speed of 0.10 m/s. Raising the minimum speed for the total product to 0.15 m/s was the only single change that classified all 31 cases correctly, and it also reduced the share of flagged cells from 0.59% to 0.54%. This value was therefore adopted. The table also shows how sensitive the results are to each parameter.

Parameter	**Values tested**	**Must be flagged**	**Must not**	**Share 3/4**
Top fraction	0.25% / **0.5%** / 1%	24 / 26 / 26	4 / 4 / 4	0.40 / 0.59 / 0.86%
Minimum difference	0.2 / 0.4 m/s	26 / 25	4 / 4	0.66 / 0.50%
$s_{\min}$, total	0.05 / **0.15** m/s	26 / 26	4 / **5**	0.66 / **0.54%**
$s_{\min}$, radials	0.03 / 0.08 m/s	26 / 24	4 / 4	0.68 / 0.49%
Angle	120° / 150°	26 / 25	4 / 4	0.52 / 0.46%
Jump at other stations	0.2 / 0.4 m/s	26 / 26	4 / 3	0.59 / 0.59%
Minimum cells	3 / 8	26 / 26	4 / 4	0.59 / 0.59%
Extension	0 / 2 cells	26 / 26	3 / 4	0.59 / 0.59%

Table 6: Sensitivity of the flag to each parameter.

With the final parameters, all 31 reference cases were classified correctly. All 26 cases that must be flagged are rated as good by the existing EU HFR Node QC, so each of them is an error that the existing QC misses and the new flag detects. The five cases that must not be flagged all remained good.

Cross-check

Table 7 shows the result of the cross-station check for the selected windows.

Product	**Windows**	**Outburst**	**Contra-dicted (4)**	**One other station (3)**	**Not checkable (3)**	**Confirmed**
AURI	510	452	39	11	8	0
PIRA	1,410	699	616	81	8	6
TRI1	2,957	2,562	303	78	11	3
IZOL	2,631	1,778	689	109	55	0
Total	3,486	2,731	755 (all flag 4)	–	–	–

Table 7: Cross-station check. "Not checkable" = no other station has data at that location.

In the radial products, most reversals were contradicted by the other stations (flag 4), and only a few were confirmed by at least two other stations (0–6 windows per station in three months). Real features seen by several stations are therefore rare compared with the reversals seen by only one station.
For the total product, the flag records which radial stations show a jump at the location of each reversal. Of the 755 reversals in the validation months, 210 had a jump only in IZOL, 130 only in TRI1, 103 only in PIRA, 93 in two stations, and 2 only in AURI, while 208 had no jump in any radial and 9 had no radial data.
At the western edge of the coverage (longitude below 13.45°E), IZOL showed a jump in 181 of the 310 total-product reversals of the whole validation set. This could support a hypothesis that the recurring artefact at the western edge of the total product originates in the far-range radial velocities of IZOL station. Figure 11 shows an example of this hypothesis: a diagonal strip of reversals in the south-western part of the field, most of them traced to IZOL.

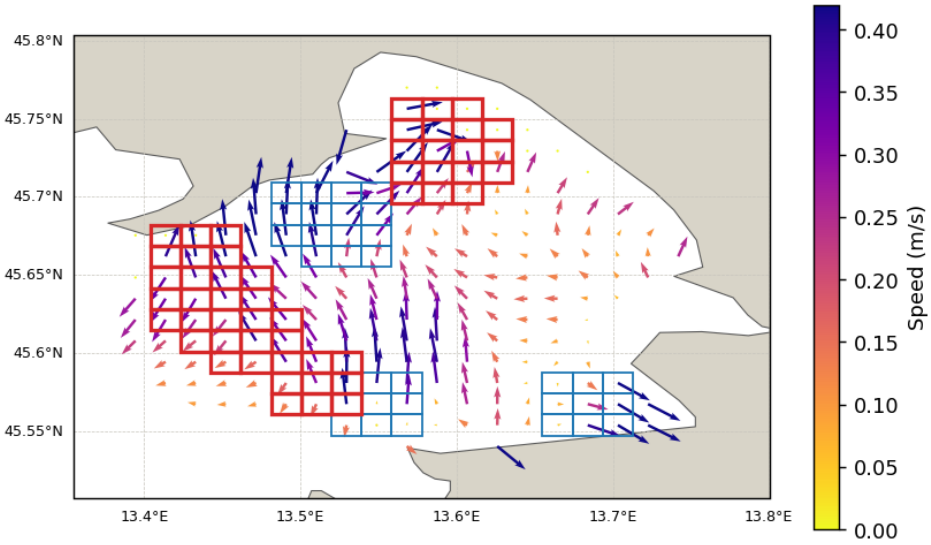


Figure 11: New QC flag in the total product, 2024-10-15. Red squares = flag 4, blue squares = outburst. The diagonal strip of reversals in the south-west was traced mainly to IZOL (8 windows), with one window traced to PIRA and two without a jump in any radial.

Figure 12 shows a radial example in which all possible results of the cross-station check occur in the same field.

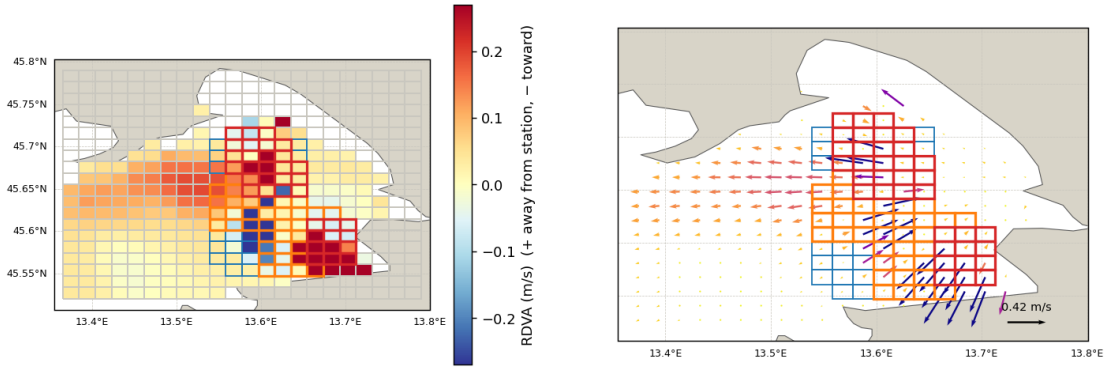


Figure 12: New QC flag for TRI1, 2024-10-26: radial velocity (left) and velocity vectors (right). Red squares = flag 4 (contradicted by the other stations), orange = flag 3 (seen by only one other station), blue = outburst or confirmed by at least two other stations. In this field 17 windows were classified as outbursts, 3 were confirmed as real, 15 were seen by one other station, and 7 were contradicted.

Comparison with the existing methods

Table 8 compares the new flag with the existing EU HFR Node QC. The existing QC evaluates single cells or whole scans (speed, variance, number of vectors, geometry, land), and it has no test for the physical consistency of neighbouring vectors. As a result, all 26 artefacts of the reference set pass as good data, while genuine Bora extremes are partly rejected by the speed threshold. The new flag detects all 26 artefacts, leaves the Bora cases and the real front as good, works on all five products, and identifies the radial station from which a total-product reversal most probably originates.

	Existing EU HFR Node QC	**New kernel QC flag**
Scope	single cells or whole scans: speed, variance, count, geometry, land	3×3 neighbourhood in all five products, plus the other stations at the same time
Unphysical divergence and shear	not detected (26 of 26 reference artefacts rated good)	26 of 26 reference artefacts detected; at most 1.25% of tested cells flagged 4
Bora extremes	partly rejected by CSPD_QC (example: 66.7% of the cells on 25 Dec 2025, 11:00)	not flagged
Real fronts	–	example: TRI1 front on 2023-11-03 confirmed by other stations
Origin of total-product errors	–	radial station(s) recorded in `KREV_SRC`

Table 8: Comparison of the new flag with the existing methods.

5 Further Considerations

5.1 Larger kernel windows

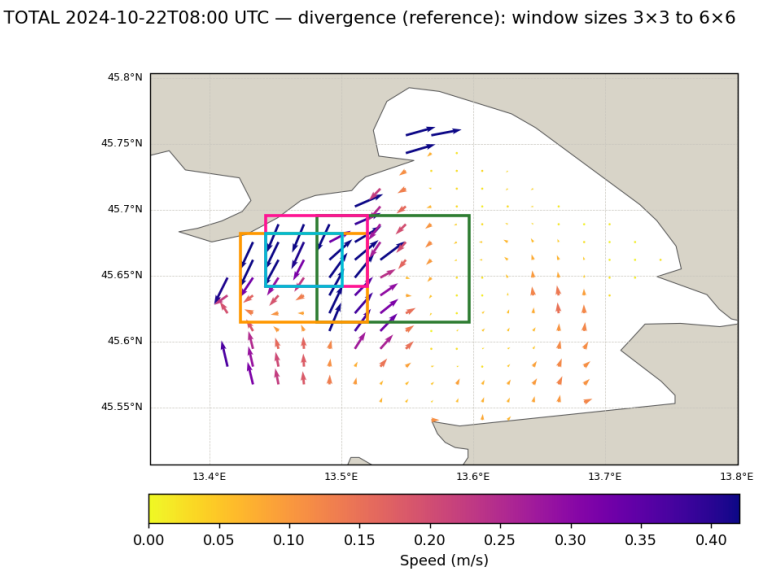
For this analysis, the existing scripts *kernel_features.py* and *kernel_clustering.py* were extended to any window size, and the script *kernel_size_compare.py* was developed. The outputs are: `kernel_sizes_coverage.csv`, `kernel_sizes_examples.csv`, `kernel_sizes_<product>_<time>.png`, `kernel_featuremaps56_<product>_<time>.png`.

A window only sees structures up to its own size. A patch of four or five cells flowing against its surroundings may be missed by a 3×3 window, because both halves of the window can fall inside the patch. The response across window sizes might also separate artefacts from real features: a one-cell artefact should be strongest in a 3×3 window and fade as it is averaged over more cells, whereas a real front or eddy should hold or grow.
To test this, the same four half-split kernels (Table 1) were computed on 5×5 and 6×6 windows. As for 3×3, odd windows have a middle row, column or diagonal with weight 0, while in even windows every cell belongs to one of the two halves. The validity rule (all cells of the window QC-good) and the Isolation Forest settings were the same as before, with one forest per product and window size.

A larger window needs more neighbouring good cells, so fewer windows are complete. The radial products lose few time steps, but the total product loses half of them at 6×6, because its coverage depends on how many stations are running and is often patchy.

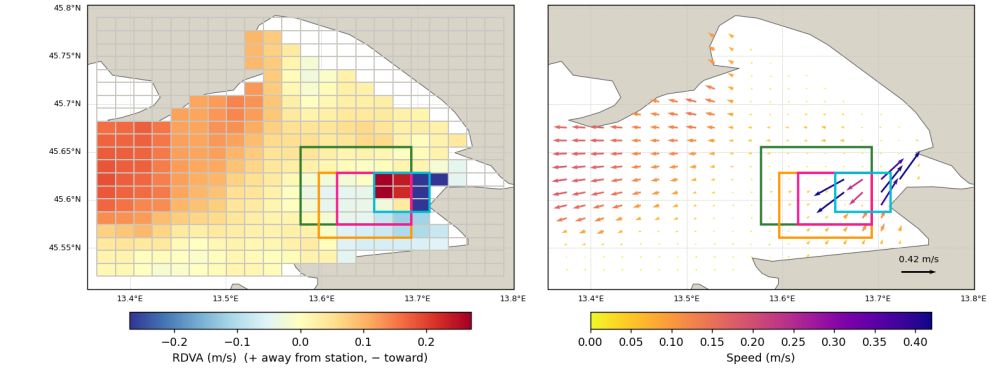
The examples were selected from the reference cases (Table 2). For each example and window size, the most anomalous window was located and its rank in the whole archive was computed. Whether the window stays on the same structure across sizes was checked cell by cell from the velocity values.
Three examples are shown in Figures 13, 14, and 15:

- Two-cell-wide structures, and long straight lines: are framed best by 4×4 or 5×5 windows (see example in the long N–S convergence line of Figure 13). The half-split kernels fit boundaries running N–S, E–W or at 45° best: larger windows hold when the boundary is long and straight along one of these cuts, and fade for structures of one or two cells or for short, irregular boundaries.
- Spikes or one-cell divergences: are framed best by 3×3 windows (see example in Figure 14).
- Wide structures: are caught entirely by the 6×6 windows (see example in Figure 15), but they can lose precision. Moreover, a real front and a wide artefact behave in the same way, and therefore distinguishing the two would still require a cross-station check.



Most anomalous window of each size at this hour (rank 1 = most anomalous window of that size, whole archive):
3×3 (cyan): score 0.767, rank 22 of 4,739,034 (top 0.00046%), box centre 45.662N 13.472E
4×4 (pink): score 0.761, rank 2 of 3,439,826 (top 0.000058%), box centre 45.669N 13.481E
5×5 (orange): score 0.738, rank 117 of 2,445,516 (top 0.0048%), box centre 45.648N 13.472E
6×6 (green): score 0.730, rank 45 of 1,675,971 (top 0.0027%), box centre 45.655N 13.539E

Figure 13: Total product, 2024-10-22 08:00 UTC: current field. Windows: 3×3 (cyan), 4×4 (pink), 5×5 (orange), 6×6 (green).



Most anomalous window of each size at this hour (rank 1 = most anomalous window of that size, whole archive):
3×3 (cyan): score 0.789, rank 5 of 3,944,137 (top 0.00013%), box centre 45.608N 13.684E
4×4 (pink): score 0.735, rank 4,035 of 3,247,450 (top 0.12%), box centre 45.601N 13.655E
5×5 (orange): score 0.709, rank 8,450 of 2,618,371 (top 0.32%), box centre 45.594N 13.645E
6×6 (green): score 0.604, rank 89,731 of 2,037,699 (top 4.4%), box centre 45.615N 13.636E

Figure 14: TRI1, 2024-10-22 03:30 UTC. Windows: 3×3 (cyan), 4×4 (pink), 5×5 (orange), 6×6 (green).

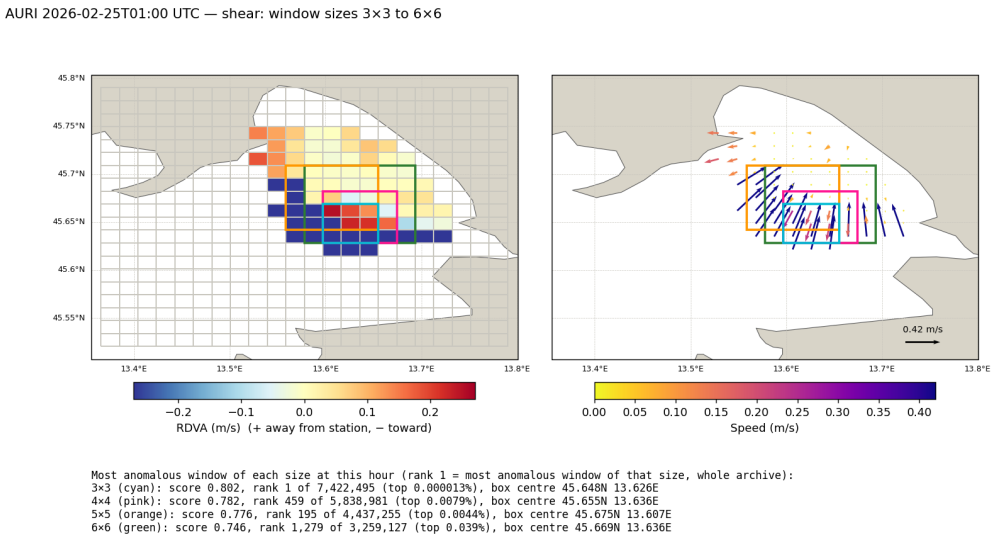


Figure 15: AURI, 2026-02-25 01:00 UTC. Windows: 3×3 (cyan), 4×4 (pink), 5×5 (orange), 6×6 (green).

No reference case was found only by the 5×5 or 6×6 windows: the change in rank with window size reflects the width of a structure, not whether it is real, so the 3×3 window remains the base window, being the only size that detects one-cell divergences. The 4×4 and 5×5 windows frame two-row bands, corners and long straight lines better and could be kept as a complement, whereas the 6×6 window frames too large an area and loses half of the total product's time steps, and is therefore not useful.

5.2 Additional kernel shapes

For this analysis, a "shapes" variant was added to the scripts *kernel_features.py* and *kernel_clustering.py*, and the script *kernel_shapes_compare.py* was developed. The outputs are: `kernel_shapes_compare.csv`, `kernel_shapes_partB.png`, `kernel_shapemaps_<product>_<time>.png`.

The half-split kernels compare two halves of a window. A single deviating cell, a thin streak or a small patch inside the window is partly diluted in the half means. Six additional shapes were therefore defined, each returning the mean of its + cells minus the mean of its − cells (Table 9 and Figure 16). The spike and the stripes were applied to 3×3 windows only, since the structures they target are one cell wide and fade in larger windows. The patch kernel needs a ring around a 2×2 centre and was therefore applied to 4×4 windows.

Kernel	+ cells minus − cells	Target
SPK (spike)	Centre cell − mean of the 8 neighbours	Single-cell spikes and reversals
STH (stripe −)	Centre row − rows to the N and S	One-cell-wide E–W streaks
STV (stripe |)	Centre column − columns to the W and E	One-cell-wide N–S streaks
STD (stripe ↘)	NW-SE diagonal − the other 6 cells	One-cell-wide diagonal streaks
STU (stripe ↗)	SW-NE diagonal − the other 6 cells	One-cell-wide diagonal streaks
PAT (patch)	Centre 2 × 2 − ring of 12 cells	Small patches inside the field

Table 9: Additional kernel shapes.

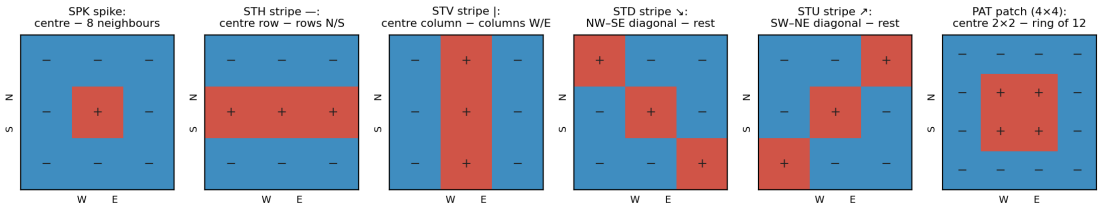


Figure 16: The additional kernel shapes. Each kernel returns the mean of the red (+) cells minus the mean of the blue (−) cells.

The new shapes were added to the four half-split kernels as further features, giving 9 kernel values per channel for the 3×3 windows and 5 for the 4×4 windows. The validity rule and the Isolation Forest settings were unchanged. The two versions were compared on the reference cases (Table 2) using the whole-archive rank of the most anomalous window that contains the error cell. Adding the six shapes to the Isolation Forest did not improve the ranking of the reference cases: most of them ranked lower than with the half-split kernels alone. The Isolation Forest scores a window as anomalous if any combination of its features is unusual. With 9 (radial) or 18 (total) features instead of 4 or 8, there are more ways for a window to be unusual, and the errors that the half-split kernels already detected are diluted among them.

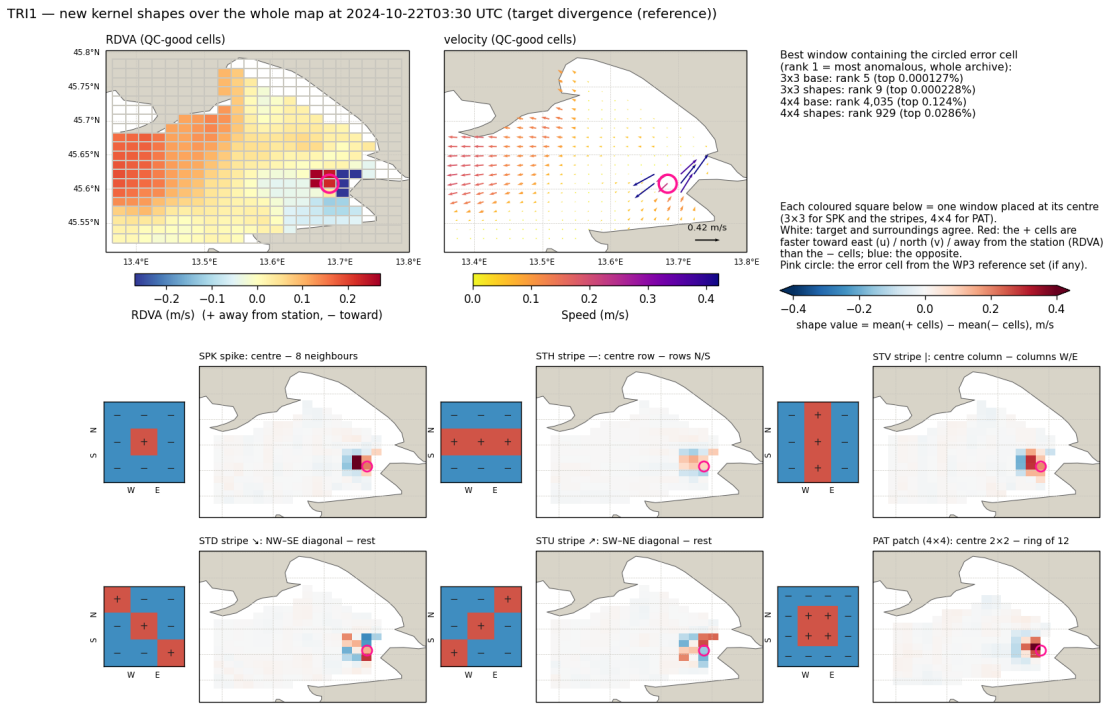


Figure 17: TRI1, 2024-10-22 03:30 UTC, single-cell spike (circled): radial velocity and current field (top), and the values of the six additional shapes over the whole map (bottom).

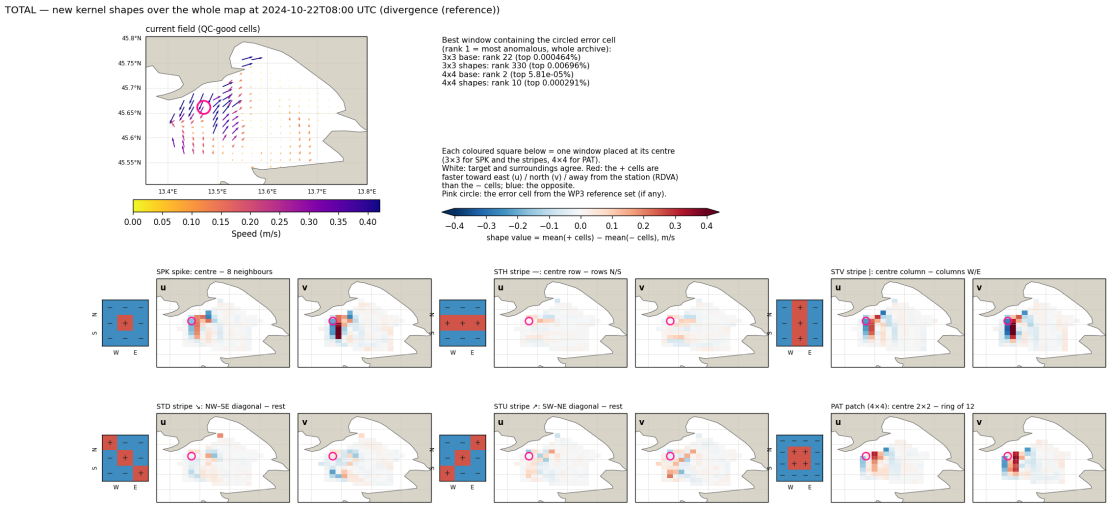


Figure 18: Total product, 2024-10-22 08:00 UTC, divergence (circled): current field (top), and the values of the six additional shapes over the whole map (bottom).

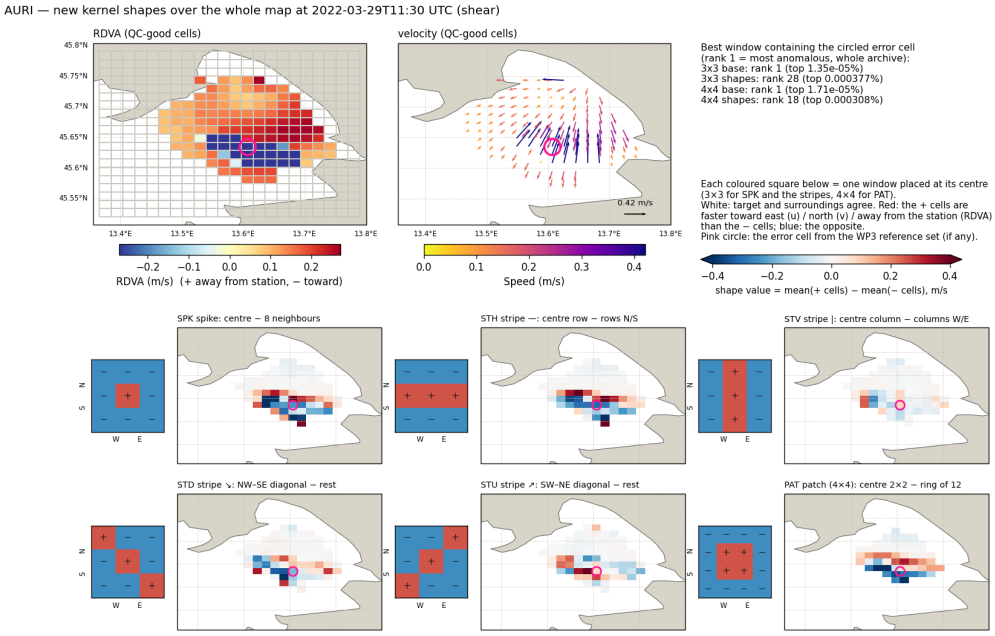


Figure 19: AURI, 2022-03-29 11:30 UTC, broad block with a shear (circled): radial velocity and current field (top), and the values of the six additional shapes over the whole map (bottom).

Considered individually, without the Isolation Forest, each shape works for some structures and not for others:

- SPK, STD, STU and PAT work well for spikes and reversals of one or two cells (Figure 17), but not for divergences over a larger area.
- STH and STV show long horizontal or vertical lines of divergence in the total product (Figure 18), but the half-split kernels are about twice as strong on the same line, because a centre column inside a symmetric divergence is compared with two sides of opposite sign.
- For broad jets and blocks (Figure 19) the shapes are of little use: inside a broad block each cell equals its neighbours, so the shapes are close to zero, and only the edges and corners of the block are highlighted, producing noisy maps.

The additional shapes are cell-scale detectors: they work well for single-cell spikes, thin streaks and small patches. They are weak for broad blocks, jets and long lines, and blind at the edge of the coverage. They therefore complement the half-split kernels rather than replace them. Mixed into one Isolation Forest, however, they dilute the ranking. A better option for WP4 is to keep them separate, having a second Isolation Forest with its own flag in the EU HFR style: `KSPK_QC` for spikes, streaks and patches next to `KREV_QC` for reversals and divergences, combined into the overall flag.

6 Conclusions

A kernel-based QC flag was developed that detects all 26 reference artefacts missed by the existing QC, leaves real Bora events and fronts as good, flags at most 1.25% of tested cells as bad, and traces total-product errors to their source radial station.

Limitations
The validation in task 4 showed the limits of the method:

- only cells inside a complete 3×3 window of good cells can be tested, so sparse areas and the edges of coverage gaps remain untested,
- a radial station cannot see a velocity jump perpendicular to its beam, so a real front may appear smooth to one station; with two confirmations required, a real front covered by only two stations remains flagged as probably bad,
- reversals in the total product are never cleared, because the radial products from which it is computed cannot confirm it independently,
- the parameters were tuned on the same reference cases on which they were tested, so the result on the reference cases is a fit rather than an independent score,
- the validation covers only the 2021-2026 period available from wp1, and some validation months are affected by station outages.

Can unsupervised algorithms separate artefacts from other situations?
The k-means clustering on whole current fields separated the data into flow regimes, but the artefacts averaged out in the cluster means, because they occur at a different place each time. Therefore the Isolation Forest algorithm was implemented. The Isolation Forest is an unsupervised method, which found the unusual windows: the most anomalous 1% contain adjacent cells with opposite radial velocities 18 to 49 times more often than average, and they include all of the reference artefacts. However, it cannot decide on its own whether an unusual window is an artefact or a real feature. Hence, unsupervised algorithms such as this are well suited to *finding* anomalies without labelled examples, but on their own they do not separate artefacts from physical situations. In the method proposed here, this separation is achieved by combining the unsupervised detection with a physically motivated rule: the classification into reversals and outbursts, and the consistency check with the other radar stations.

7 Additional Remarks

The script *persistent_bad_cells.py* was developed, alongside the tables *persistent_bad_cells.csv*, *persistent_bad_cells_by_period.csv*, and figures *persistent_bad_cells_map.png*, *persistent_bad_cells_by_period.png*.

Throughout the analyses, for both this study and wp2, some cells inside the domain have almost always, if not always, been flagged as bad by the existing QC flags.
This evaluation pinpoints the cells that are (almost) always unusable, for every product (Total and each radial station), excluding the outages.
In order to examine the cell issues, the cells have been divided into 3 categories (Table 10):

Category	Rule
A – almost always bad	Data flagged as QC-bad $> 90\%$
B – often bad	Data flagged as QC-bad 50–90%
C – rarely usable	Existing QC-good data in $< 1\%$ of running hours

Table 10: Categorisation of persistently QC-bad cells in the GOT.

Figures 20, and 21 depict the results. In both cases, each grid cell is split into 5 strips, one strip for each product (Total, AURI, PIRA, TRI1, IZOL). Hence, it is possible to see the position of the persistent bad cell and which product causes that error. The station triangles use the coordinates from wp1.

Figure 20 represents persistently bad cells for the whole time series 2021-2026:

- **A**: at 45.54°N 13.645°E (close to Izola) due to stations AURI and TRI1 (flagged by the OWTR_QC); at 45.648°N 13.742°E (Trieste) due to stations TRI1 and IZOL (flagged by the OWTR_QC flag); at 45.702°N 13.684°E due to the Total product.
- **B**: a N–S line of Total cells between stations AURI and IZOL (flagged by the GDOP_QC); at 45.54°N 13.587°E (close to PIRA) due to the Total product (flagged by the GDOP_QC).
- **C**: 9 Total and 4 PIRA cells along the northern coast.

Figure 21 is divided into two subplots corresponding to different periods of station deployment. The left subplot shows the period from 2021 to 2023, when only two stations (AURI and PIRA) were operational, while the right subplot covers the period from 2023 to 2026, when all stations were installed and operational:

- **Period 1** (only AURI + PIRA):
  Total has 17 A cells (flagged by GDOP_QC) in a block (next to AURI), 42 B cells over most of the central gulf and 24 C cells (edge of the 2-station coverage in the far west and north coast); AURI has one A cell, close to Izola; PIRA has 3 north-far-range C cells.

- **Period 2** (all four installed): the GDOP block is gone. What remains: the land-mask cells close to Izola (caused by AURI, TRI1) and Trieste (caused by TRI1, IZOL, and PIRA); one Total B cell at 45.594°N 13.703°E; the north-coast C cells.

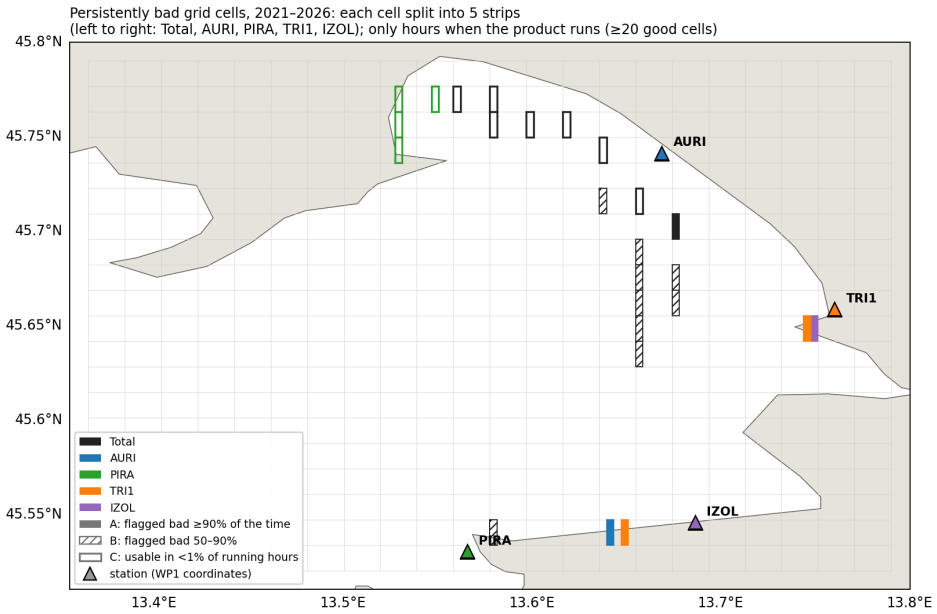


Figure 20: Map of the cells persistently flagged as bad by the existing QC algorithm for the time series 2021-2026.

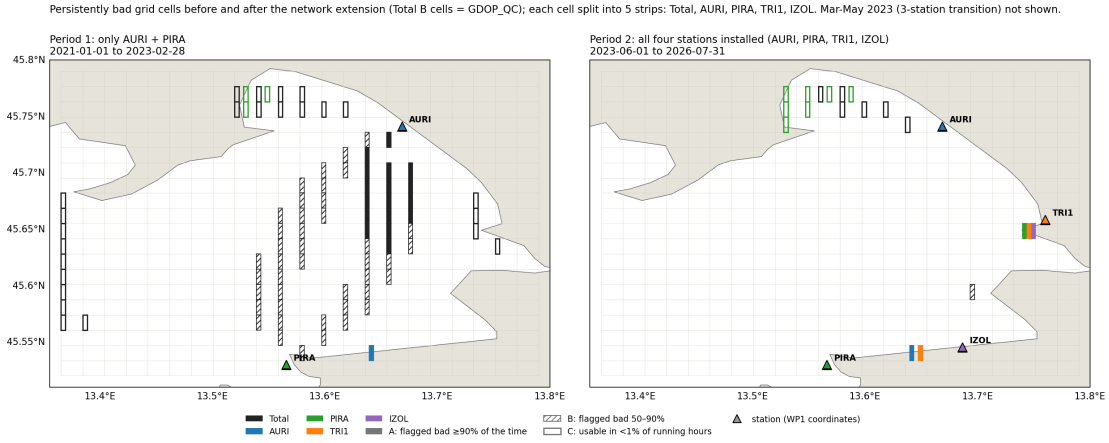


Figure 21: Map of the cells persistently flagged as bad by the existing QC algorithm divided by period.